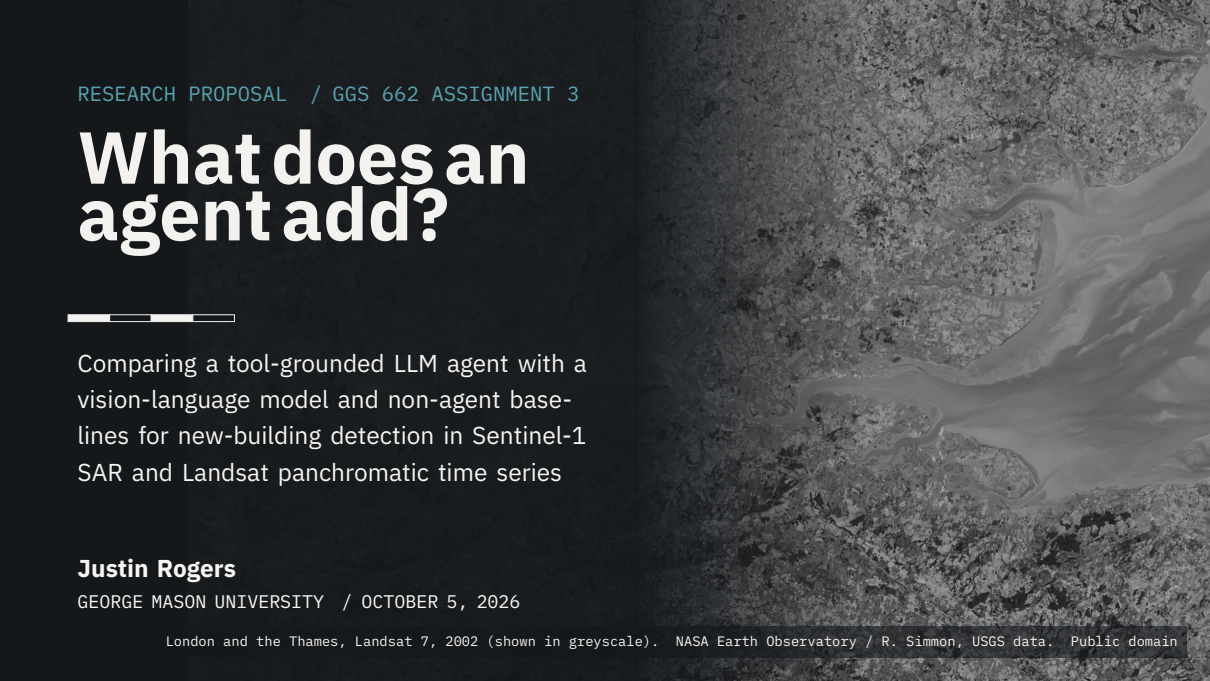
RESEARCH PROPOSAL / GGS 662 ASSIGNMENT 3

What does an agent add?

Comparing a tool-grounded LLM agent with a vision-language model and non-agent baselines for new-building detection in Sentinel-1 SAR and Landsat panchromatic time series

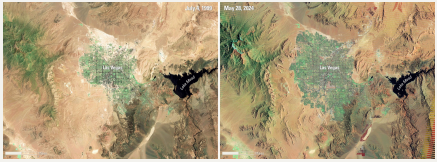
Justin Rogers
GEORGE MASON UNIVERSITY / OCTOBER 5, 2026



London and the Thames, Landsat 7, 2002 (shown in greyscale). NASA Earth Observatory / R. Simmon, USGS data. Public domain

The problem: large image archives, unreliable AI interpreters

- Satellite archives now hold years of monthly SAR and optical images per place. Reading change from them by hand does not scale.
- Current multimodal models are weak at this:
  - best of 18 models: 34.1% **accuracy** (share of city-change benchmark questions answered correctly) (Xuan et al., 2025)
  - LLM agents on yes/no Earth-observation questions: 40% **accuracy** zero-shot, about 60% with self-reflection, where guessing scores 50%; generated code **failed to run** in over 44% of runs (Kao et al., 2026)
  - TEOChat, a temporal vision-language model, handles at most 8 images per sequence (Irvin et al., 2025)
- Tool-using agents are the promising route (Feng et al., 2026), but SAR and panchromatic data are rarely tested.

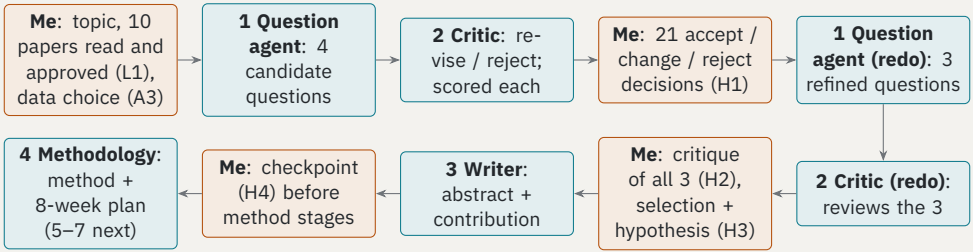


Las Vegas 1999 vs 2024 (Landsat 7, USGS; public domain)

What this study tests

Whether grounding an LLM in sensor tools makes it more accurate and less prone to fabrication than a VLM reading the same imagery, and what its reasoning adds over non-agent rules given the same tool outputs.

How I used the multi-agent workflow



Three refined questions and my critique

Question	Scientific value (my critique)	Feasibility in 8 weeks (my critique)	Origin
Q1. Is a tool-using agent *no worse* than a fixed pipeline with the same tools, better where the tools disagree, and does it abstain when data are missing?	Strong comparison. It is detection, not description, but tool outputs from classic CV methods act as extra features instead of the model interpreting from scratch.	Feasible at test level; a production system across many areas would take longer.	AI, edited by me
Q2. Does panchromatic imagery (15 m resolution) add anything to Sentinel-1 SAR?	Good idea, but I would flip it: panchromatic supported by SAR, since panchromatic is more common in commercial imagery.	Hard from scratch in 8 weeks; possible with an AI expert or by building on prior research.	AI, edited by me
Q3. Does neighbouring-cell context help the agent place change in the right cell?	Reasonable, but too general: extra context could be almost anything, and it has no target problem.	Feasible.	AI

Selected question: Q1, with a VLM comparison

Research question

In held-out SpaceNet 7 areas (2017–2020), does an LLM agent with built-in SAR and panchromatic tools detect new buildings (presence and month) in 100 m cells more accurately than a vision-language model that looks at the imagery directly, and than non-agent baselines given the same tool outputs? Does it fabricate fewer answers?

Why it is the strongest

- Answers my core interest: VLMs vs tool-grounded agents.
- Same-tool baselines isolate what the agent's reasoning adds.
- Feasible at test level with existing labels; Q2 becomes a pipeline-first secondary; Q3 deferred.

Hypothesis

- Tool grounding reduces hallucination: the agent is more accurate than the VLM and fabricates fewer answers.
- Against a tuned rule and a fitted rule on the same tool outputs, the agent is at least as accurate (difference with a confidence interval) and better where the tools disagree.

Exactly the same inputs for the LLM agent and the VLM: the same 1 km image chips (panchromatic at 15 m resolution, SAR at 10 m resolution), the same 8 dates, the 100 m square outlined, and the same question.

AI suggestions I accepted, changed or rejected

AI suggestion	Decision	Why
Critic: an agent-vs-VLM comparison is confounded; first log dropped the VLM	**Changed**	**Reinstated by me**: it is my core question. Made fair with 1 km chips and the VLM's full frame limit, and paired with same-tool baselines.
Hand-label 300–500 cells; use NAIP for dates	**Rejected**	SpaceNet 7 already has monthly building labels from separate imagery; saves about two weeks.
Critic: add a fitted baseline; replace the margin δ with confidence intervals	**Accepted**	A second baseline on the same tool outputs; intervals instead of an arbitrary margin.
Critic's revised question: agent vs fixed pipeline	**Accepted**	The only contrast that isolates what the agent adds.
Panchromatic as an optional ablation	**Changed**	Required secondary analysis, and panchromatic-led (my Q2 critique).
Single area split into blocks	**Changed**	Whole areas held out (AOI-level split): less leakage, better external validity.
Critic: reject the neighbourhood question (Q3)	**Changed**	Not rejected, just not selected: reasonable and feasible, but too general for this paper; kept as a follow-on.

Full log: 21 decisions (form field H1); earlier outputs and every agent's input packet are archived.

Literature and contribution

Paper	What it established	What this work adds
Kao et al. (2026), ACL Findings	LLM agents reach about 60% accuracy at best on yes/no EO questions (guessing: 50%); generated code often fails to run	Predefined, deterministic tools instead of free-form code
Feng et al. (2026), ICLR	Tool-using EO agent (Earth-Agent, 104 tools)	A SAR + panchromatic tool set and a same-tools baseline
Xuan et al. (2025), NeurIPS	Best of 18 multimodal models: 34.1% accuracy (questions answered correctly) on a city-change benchmark	Grounds answers in time-series statistics
Lee et al. (2026), IEEE JSTARS	SAR time-series change types; up-steps linked to construction	Uses the SAR change test inside an agent
Qu et al. (2026), Remote Sens. Environ.	Dense time-series change detection; AOI-level split; misalignment creates false change	Same split logic and misregistration checks
Zuo et al. (2026), IEEE JSTARS	Hallucinated change is a known failure; change-vector-analysis calibration plus self-consistency reduces it	Measures abstention and fabricated answers

Proposed contribution: a controlled comparison of a tool-grounded LLM agent with a vision-language model and with same-tool baselines, on a dense SAR + panchromatic archive, scored against independent monthly labels. Proposed, not established.

Data and method

Data (all public)

- **Panchromatic:** Landsat-8 band 8 at 15 m resolution (each pixel covers 15 × 15 m of ground), monthly, already co-registered (MuRA-T, 60 SpaceNet 7 areas, 2017–2020)
- **SAR:** Sentinel-1 C-band GRD, one orbit track per area, co-registered by me
- **Reference:** SpaceNet 7 monthly building footprints, traced on separate PlanetScope imagery
- **Unit:** 100 m cells in 10–20 areas; whole areas held out for testing

Steps

1. Build deterministic tools: SAR series and likelihood-ratio change test; normalised panchromatic series and change magnitude; a pretrained CNN change detector; data availability
2. **Tuned baseline:** fixed pipeline with the same tools, tuned on development areas
3. **Agent:** same tools, answers `presence / onset month / abstain`
4. **VLM:** reads the same 1 km chips on the same 8 dates as the agent's tools
5. **Fitted baseline:** simple classifier on the same tool outputs
6. Controls: same LLM without tools; panchromatic-only vs panchromatic + SAR (pipeline, all cells)
7. Freeze, pre-register, run test areas once

AI-assisted coding: Claude Code writes the download, chip, tool, harness and analysis code; I review every function and test it on synthetic images with known change. No LLM scores the answers.

Validation, spatial assumptions and risks

Validation checks

- Difference in F1 score (balance of precision and recall), plus precision and recall, for agent vs VLM and agent vs each baseline, with area-level cluster bootstrap confidence intervals
- Accuracy on cells where SAR and panchromatic disagree
- Abstention and fabricated-answer counts on unanswerable queries
- Robustness: 200 m cells, one-pixel shift, look-direction tolerance

Spatial assumptions

- Sub-pixel co-registration holds; cell size set in metres (grid may be Web Mercator)
- Labels mark when a building is visible, not when construction starts

What would weaken the claim

- The VLM matches the agent once given fair input
- A fitted rule on the same tool outputs matches the agent
- The no-tools model scores close to the agent (answers from memory)
- The agent almost never disagrees with the pipeline (then switch to the panchromatic question)
- Few test areas give wide CIs

Key risks: SAR layover moves signal between cells; Level-1 panchromatic radiometry adds false change; API budget.

Eight-week plan, journal and co-authors

Week	Work
1	Download data; choose areas, VLM and CNN; draft pre-registration
2	Grid, labels, 1 km chips; validate each tool on its own
3	Agent and VLM harness; tune and fit baselines; pilot
4	Freeze, pre-register; run all arms on test areas
5	Score; area-level confidence intervals; robustness
6	Draft paper
7	Co-author and critic review; check claims; revise
8	Final checks; package code and data; submit

Journal: *International Journal of Applied Earth Observation and Geoinformation* (Elsevier, open access). **2025 JIF 8.2**, Q1 in Remote Sensing (7/66); source: Clarivate Journal Citation Reports, 2025 JCR year (June 2026 release). Publishes applied EO methods, SAR and building change detection, and GeoAI, which fits an agent-vs-VLM evaluation on SAR + panchromatic time series. **Possible co-authors** (neither has agreed)

- **Dr. Edward Oughton**, GMU Geography & Geoinformation Science: agentic GeoAI and spatial computing. Checks: workflow design, evaluation protocol, reproducibility.
- **Dr. Konrad Wessels**, GMU Geography & Geoinformation Science: SAR and time-series remote sensing. Checks: SAR preprocessing, co-registration, physical plausibility of SAR change.

Sources and AI disclaimer

Data: MuRA-T (Purdue RVL; Deshmukh et al., 2023, arXiv:2302.12301, GPL v3); SpaceNet 7 (Van Etten et al., CVPR 2021, CC BY-SA 4.0); Copernicus Sentinel-1 (ESA); Landsat-8 (USGS). **Images:** title, NASA Earth Observatory image by Robert Simmon using USGS Landsat data (public domain, shown in greyscale); slide 2, USGS Landsat Missions, Las Vegas 1999–2024 (public domain). Full references: supporting slides.

AI tools and what they did: Claude Code (Anthropic, Claude Opus 5.5) built the orchestration script and prompt packets, ran the question, critic, writer and methodology roles as fresh agents (the methodology-critic, data + tools and validation roles were not run before submission), searched the literature and checked DOIs against Crossref, found the MuRA-T dataset and openly licensed images, checked the journal metric in JCR through my GMU login (the first choice, IJDE, was on hold, so the target became JAG), drafted the first rows of the decision log (H1) and the data notes at my direction, and generated these slides.

My own checks: read and approved all 10 references; wrote my own critique of the three questions; selected the question and hypothesis; chose the co-authors; added my own decision-log rows after the critic's review, including reinstating the VLM comparison. Agreement between agents was not treated as validation.

Questions?

WHAT DOES THE AGENT ADD THAT THE SAME TOOLS DO NOT?

SUPPORTING

References (APA 7), 1 of 2

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SUPPORTING

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All 12 full PDFs are saved in the project folder; every claim on these slides was checked against them (`literature/claim_check.md`). Deshmukh et al. (2023) is a preprint; Lee et al. (2026) is an accepted manuscript.

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Detailed eight-week plan

Week	Tasks	Deliverable	Main risk
1	Download MuRA-T, SpaceNet 7, Sentinel-1; choose areas; choose the VLM and CNN; draft pre-registration	Area inventory and split; pre-registration draft	Too little Sentinel-1 or panchromatic coverage
2	100 m grid in metres; rasterise labels; pick the 8 dates; render 1 km chips; validate each tool on its own	Chip set; tool library; tool-validation report	Change test or CNN fails at 15 m resolution
3	Agent and VLM harness; tune the rule and fit the second baseline; full development pilot; set sample size and API cap	Pilot results; frozen prompts and rules	VLM setup overruns; too few conflict cells
4	Freeze and pre-register; run agent, VLM, controls and both baselines on the test areas	Pre-registration; raw outputs and logs	API cost or rate limits
5	Score; area-level bootstrap confidence intervals; robustness (shift, 200 m cells, layover)	Results tables and figures	Wide intervals with few test areas
6	Draft methods, results, discussion, limitations	Full draft	Over-claiming
7	Critic and co-author review; check every claim against results and references;	Revised draft; claim audit	Review finds an analysis error

SUPPORTING

An agent output and my revision

Agent (stage 1 redo), Q-B

“Does giving a tool-using LLM agent Landsat-8 panchromatic tools, in addition to Sentinel-1 tools, change how accurately it identifies 100 m cells that gain new buildings?”

Critic (stage 2) and the first decision log

The critic called an agent-vs-VLM comparison confounded; the first decision log (drafted by Claude) dropped the VLM arm.

My revision

Flip the direction: panchromatic is the lead sensor, SAR supports it, because panchromatic is more common in commercial imagery. Kept as a required secondary analysis inside Q1.

My decision

Reinstated the VLM comparison, my core question, and addressed the critique: the VLM gets readable 1 km chips and its full frame limit, and same-tool baselines isolate the agent’s reasoning.

Outputs: `outputs/research_questions.md`, `outputs/question_review.md`; earlier versions and every prompt packet in `outputs/history/` and `outputs/packets_used/`.